Section	Task/ Component	Detailed Criteria	Points
Section 1: Knowledge Base Preparation	**Manual Chunking Function (chunking)**	• Correctly splits kb_text using newline characters (\n). • Successfully filters out empty strings or strings containing only whitespace. • Output is a correct list of chunks.	**1**
Section 2: Embed Chunks (Setup)	**Embedding Model Initialization**	• Correctly initializes SentenceTransformer.	**0.5**
	Chunk Embedding	• Successfully encodes the list of chunks into a numpy or PyTorch vector matrix (chunk_embeddings). • The resulting shape and data types are correct for subsequent operations.	**1**
Section 3: Calculate Similarity (Core Logic)	**Cosine Similarity Function (cosine_sim)**	• **Core Formula:** Correctly implements cosine similarity using numpy operations (dot product and norms), accurately reflecting the mathematical formula. • **Edge Case Handling:** Includes logic to safely handle vectors with zero norm (e.g., returning 0), preventing division by zero errors.	**0.5**
Section 4: Manual Retrieval	**Query Embedding**	• Correctly embeds the input query string.	**0.5**
	Similarity Calculation/ Iteration	• Correctly iterates through all chunk_embeddings and calls the custom cosine_sim function for comparison.	**1**
	Top-K Selection Logic	• Correctly uses numpy.argsort() (or equivalent indexing/sorting method) to find the indices corresponding to the **highest** top_k similarity scores.	**1**
Section 5: LLM	**LLM Loading**	• Successfully loads the AutoTokenizer and AutoModelForCausalLM.	**0.5**

Section	Task/ Component	Detailed Criteria	Points
Loading (Setup)			
Section 6: Generation	**Raw LLM Generation (raw_generate)**	• Implements a simple, non-contextual prompt (only system instruction + query).	**0.5**
	RAG Function Integration	• Correctly integrates and calls the retrieve(query, top_k=3) function to fetch the context.	**0.5**
	RAG Prompt Construction	• **Crucial:** Formats a structured prompt where the retrieved context is clearly injected **before** the question, guiding the LLM to rely on the external information.	**1**
Section 7: Testing & Analysis	**Output Verification**	• Test execution correctly displays: the input query, the list of **Top K** retrieved chunks, their **index**, and their **calculated similarity score** (e.g., score=0.735).	**0.5**
	Analysis 1: RAG vs. Raw Comparison	• Provides a brief, clear analysis comparing the two outputs, explicitly discussing how the RAG answer is factually grounded in the provided text, while the Raw LLM output may be incorrect or ungrounded (hallucination/general knowledge).	**1**
	Analysis 2: Outcome Comparison	• Attempts the required analysis (e.g., comparing outcomes by different prompts, varying top_k value, or discussing the quality of the answer compared to the source text).	**0.5**
TOTAL			**10**